Lect# 5: Project Time Management (PTM)

Project Time Management

- Project time management involves the processes required to ensure timely completion of a project.
- Processes include:
  1. Activity definition
  2. Activity sequencing
  3. Activity duration estimating
  4. Schedule development
  5. Schedule control

1. Activities Definition Process

- Involved identifying specific activities that pjt team members & stakeholders must perform to produce the pjt deliverables.
- An activity is element of work, normally found on WBS, that has duration, cost & resources requirements.

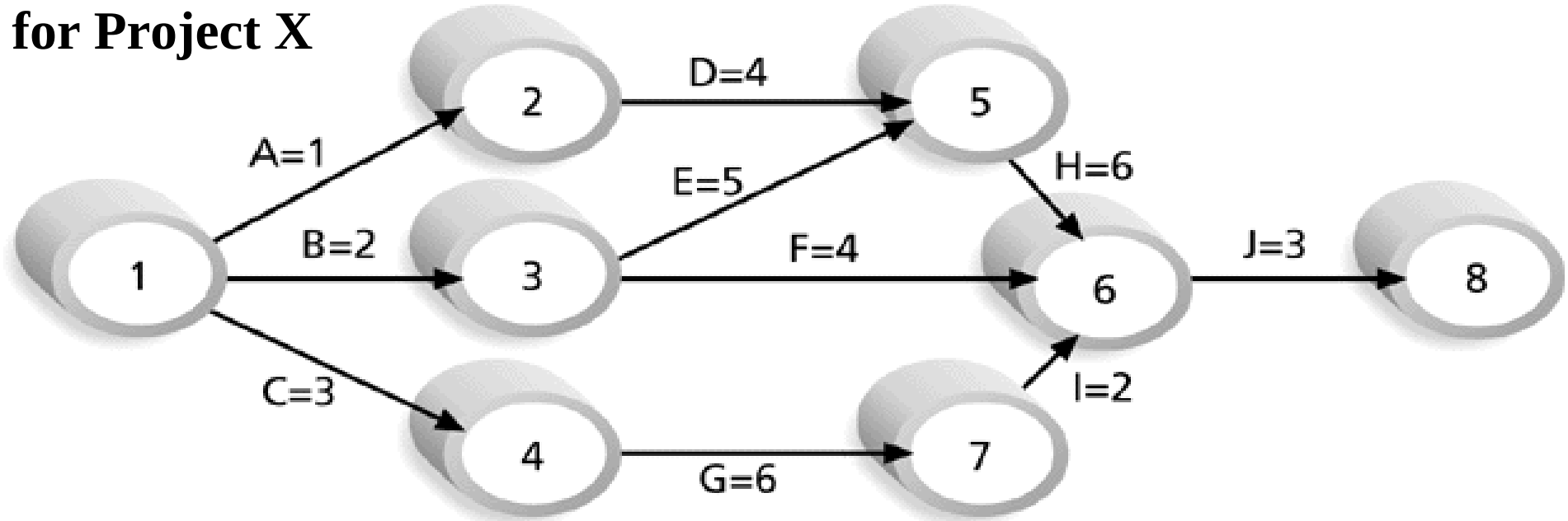
2. Activity Sequencing

- Involves reviewing the activities in the detailed WBS, detailed product descriptions, assumptions n constraint to determine the relationship bet activities, and evaluate reasons for dependencies.
- Ex, does an activity have to finished before another one can start?
- Ex, can several activities done in parallel? Can some overlap?
- Determine these relationship/dependence betw activities has significant impact PM.

Tools : Network Diagrams

- Project network diagrams are the preferred technique for showing activity sequencing
- A project network diagram is a schematic display of the logical relationships among, or sequencing of, project activities

Figure 5-2. Sample Activity-on-Arrow (AOA) Network Diagram for Project X



Note: Assume all durations are in days; A=1 means Activity A has a duration of 1 day.

Process for Creating AOA Diagrams

1. Find all of the activities that start at node 1. Draw their finish nodes and draw arrows between node 1 and those finish nodes. Put the activity letter or name and duration estimate on the associated arrow
2. Continue drawing the network diagram, working from left to right. Look for bursts and merges. *Bursts* occur when a single node is followed by two or more activities. A *merge* occurs when two or more nodes precede a single node
3. Continue drawing the project network diagram until all activities are included on the diagram that have dependencies
4. As a rule of thumb, all arrowheads should face toward the right, and no arrows should cross on an AOA network diagram

3. Activity Duration Estimating

- Duration includes the actual amount of time worked on an activity *plus* elapsed time
- Output;
  - Estimate duration(with doc) for each activity
  - Updates to the WBS

4. Schedule Development(SD)

- SD uses results of the other time management processes to determine the start and end date of the project and its activities
- SD goal: to create a realistic *project schedule* that provides a basis for monitoring project progress
- Important tools and techniques include Gantt charts, PERT analysis, and critical path analysis

Gantt Charts

- Gantt charts provide a standard format for displaying project schedule information by listing project activities and their corresponding start and finish dates in a calendar format
- Symbols include:
  - A black diamond: milestones or significant events on a project with zero duration
  - Thick black bars: summary tasks
  - Lighter horizontal bars: tasks
  - Arrows: dependencies between tasks

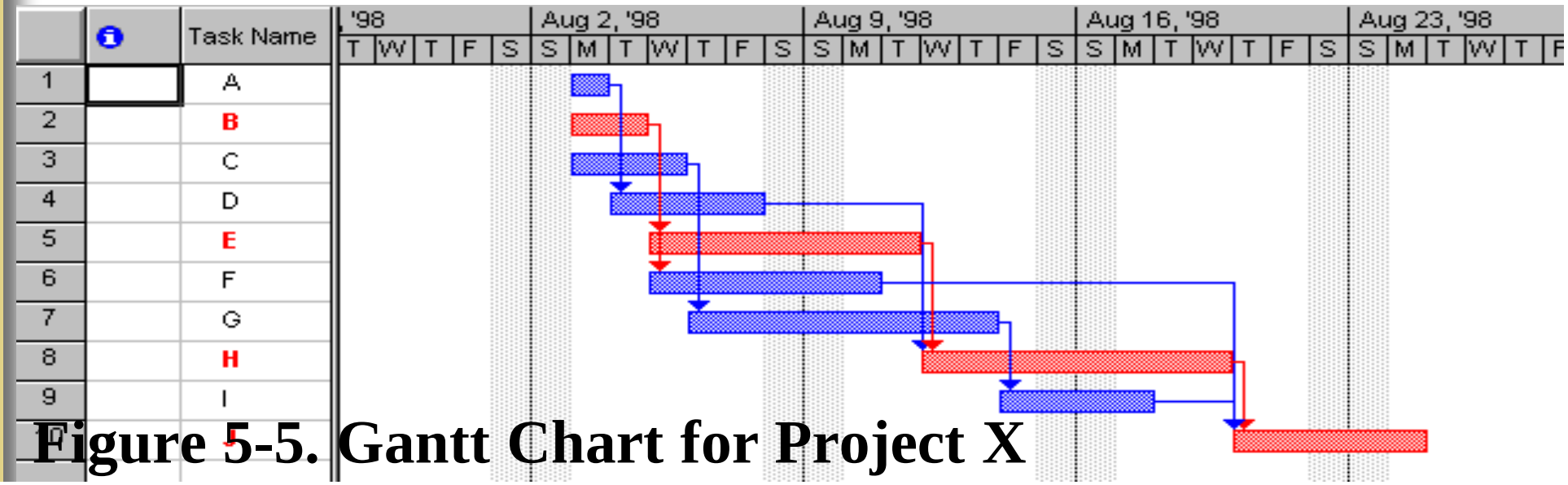


Figure 5-5. Gantt Chart for Project X

Figure 5-6. Gantt Chart for Software Launch Project

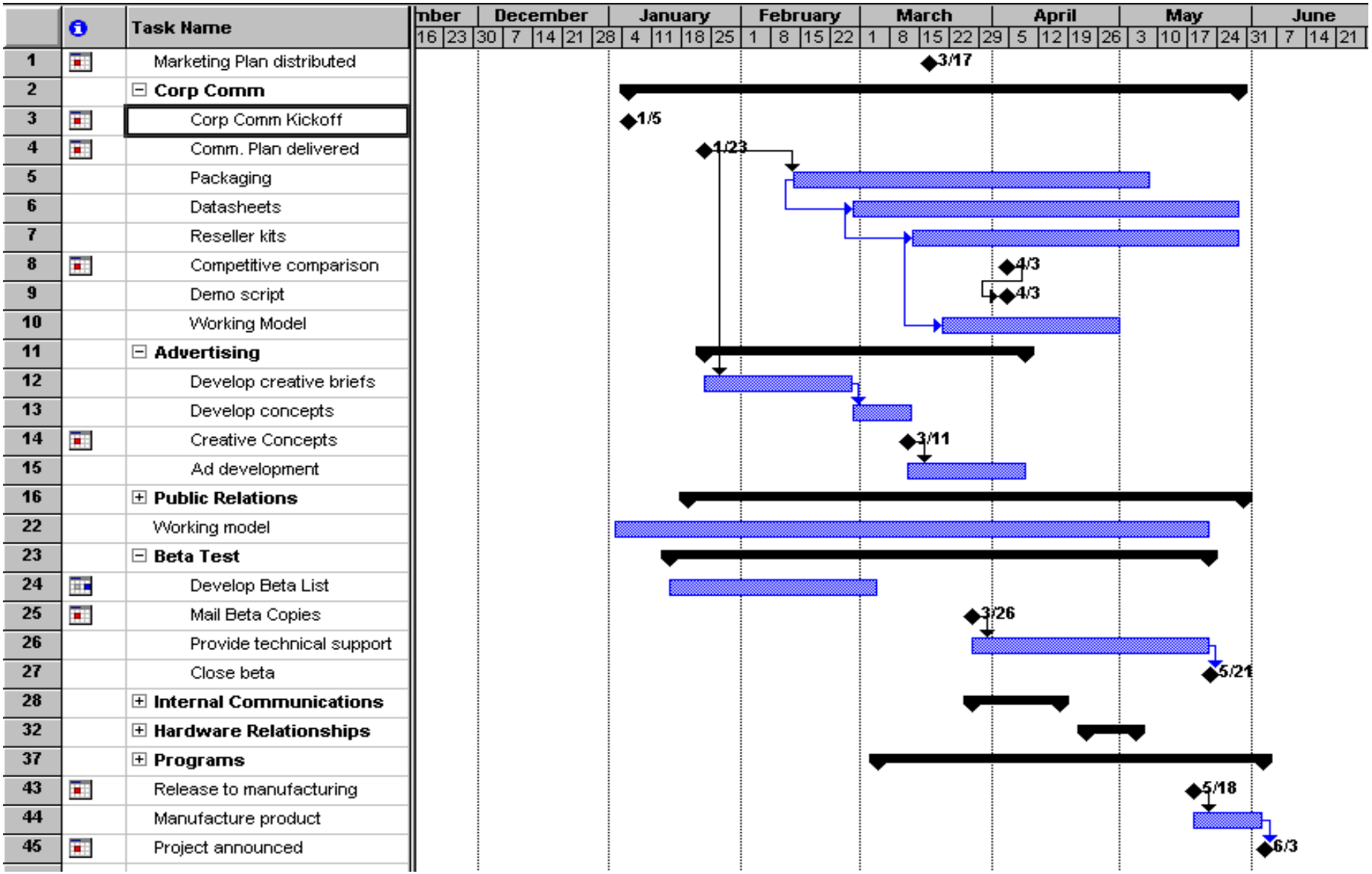


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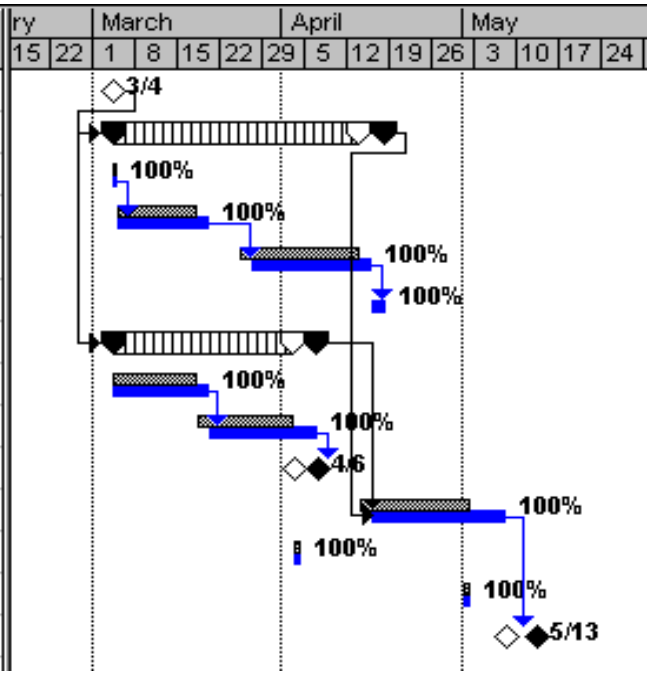
Figure 5-7. Sample Tracking Gantt Chart

	Task Name	Start	Finish	Baseline Start	Baseline Finish
1	Contract Awarded	Wed 3/4/98	Wed 3/4/98	Wed 3/4/98	Wed 3/4/98
2	**Main Task 1**	**Wed 3/4/98**	**Fri 4/17/98**	**Wed 3/4/98**	**Mon 4/13/98**
3	Subtask 1.1	Wed 3/4/98	Wed 3/4/98	Wed 3/4/98	Wed 3/4/98
4	Subtask 1.2	Thu 3/5/98	Thu 3/19/98	Thu 3/5/98	Tue 3/17/98
5	Subtask 1.3	Fri 3/27/98	Wed 4/15/98	Wed 3/25/98	Mon 4/13/98
6	Deliverable 1	Wed 4/15/98	Fri 4/17/98	Mon 4/13/98	Mon 4/13/98
7	**Main Task 2**	**Wed 3/4/98**	**Mon 4/6/98**	**Wed 3/4/98**	**Thu 4/2/98**
8	Subtask 2.1	Wed 3/4/98	Thu 3/19/98	Wed 3/4/98	Tue 3/17/98
9	Subtask 2.2	Fri 3/20/98	Mon 4/6/98	Wed 3/18/98	Thu 4/2/98
10	Deliverable 2	Mon 4/6/98	Mon 4/6/98	Thu 4/2/98	Thu 4/2/98
11	Main Task 3	Thu 4/16/98	Thu 5/7/98	Tue 4/14/98	Fri 5/1/98
12	Project Review 1	Fri 4/3/98	Fri 4/3/98	Fri 4/3/98	Fri 4/3/98
13	Project Review 2	Fri 5/1/98	Fri 5/1/98	Fri 5/1/98	Fri 5/1/98
14	Final Report and Presentation	Wed 5/13/98	Wed 5/13/98	Fri 5/8/98	Fri 5/8/98



white diamond: slipped milestone
two bars: planned and actual times

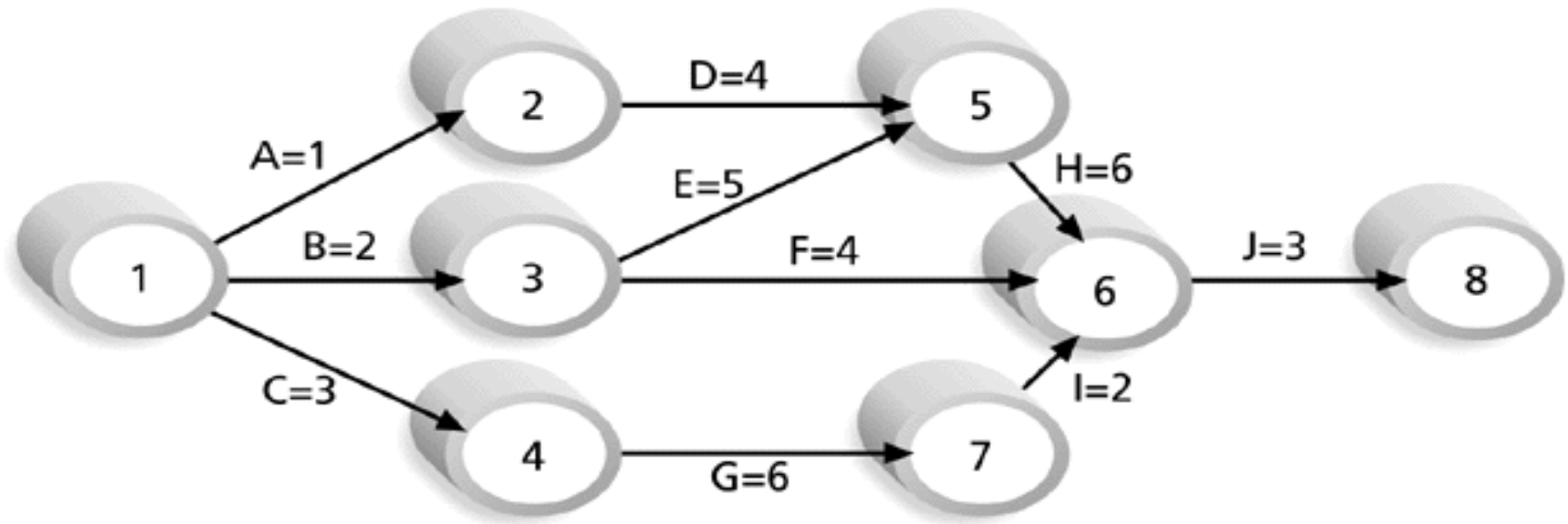
Project 98 file

Critical Path Method (CPM)

- CPM is a project network analysis technique used to predict total project duration
- A critical path for a project is the series of activities that determines the *earliest time* by which the project can be completed
- The critical path is the *longest path* through the network diagram and has the least amount of slack or float

- Finding the Critical Path
  - First develop a good project network diagram
  - Add the durations for all activities on each path through the project network diagram
  - The longest path is the critical path

Figure 5-8. Determining the Critical Path for Project X



Note: Assume all durations are in days.

Path 1:	A-D-H-J	Length = 1+4+6+3 = 14 days
Path 2:	**B-E-H-J**	**Length = 2+5+6+3 = 16 days**
Path 3:	B-F-J	Length = 2+4+3 = 9 days
Path 4:	C-G-I-J	Length = 3+6+2+3 = 14 days

Since the critical path is the longest path through the network diagram, Path 2, B-E-H-J, is the critical path for Project X.

More on the Critical Path

- If one of more activities on the critical path takes longer than planned, the whole project schedule will slip *unless* corrective action is taken
- Misconceptions:
  - The critical path is not the one with all the critical activities; it only accounts for time
  - There can be more than one critical path if the lengths of two or more paths are the same
  - The critical path can change as the project progresses

Using Critical Path Analysis to Make Schedule Trade-offs

- Knowing the critical path helps you make schedule trade-offs
- *Free slack or free float* is the amount of time an activity can be delayed without delaying the early start of any immediately following activities
- *Total slack or total float* is the amount of time an activity may be delayed from its early start without delaying the planned project finish date

5. Controlling Changes to the Project Schedule

- Perform reality checks on schedules
- Allow for contingencies
- Don’t plan for everyone to work at 100% capacity all the time
- Hold progress meetings with stakeholders and be clear and honest in communicating schedule issues

Working with People Issues

- Strong leadership helps projects succeed more than good PERT charts
- Project managers should use
  - empowerment
  - incentives
  - discipline
  - negotiation

Using Software to Assist in Time Management

- Software for facilitating communications helps people exchange schedule-related information
- Decision support models help analyze trade-offs that can be made
- Project management software can help in various time management areas

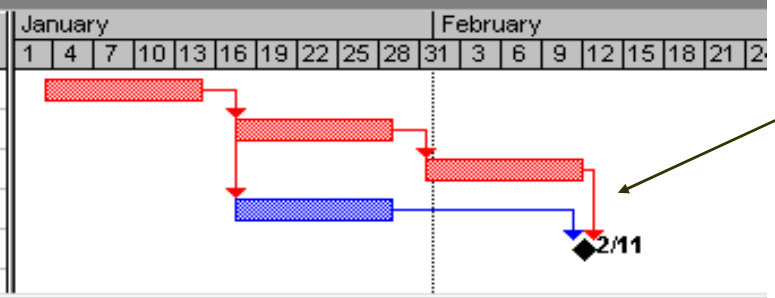
Techniques for Shortening a Project Schedule

- **Shortening durations of critical tasks by adding more resources or changing their scope**
- ***Crashing* tasks by obtaining the greatest amount of schedule compression for the least incremental cost**
- ***Fast tracking* tasks by doing them in parallel or overlapping them**

Shortening Project Schedules

compress1.mpp

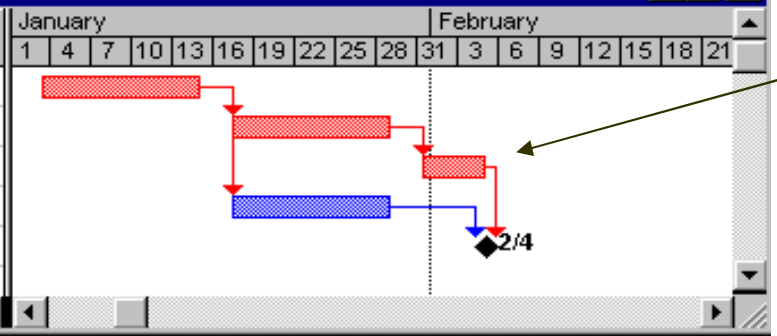
	Task Name	Duration	Start	Finish
1	A	10 days	Mon 1/3/00	Fri 1/14/00
2	B	10 days	Mon 1/17/00	Fri 1/28/00
3	C	10 days	Mon 1/31/00	Fri 2/11/00
4	D	10 days	Mon 1/17/00	Fri 1/28/00
5	E	0 days	Fri 2/11/00	Fri 2/11/00



Original schedule

compress2.mpp

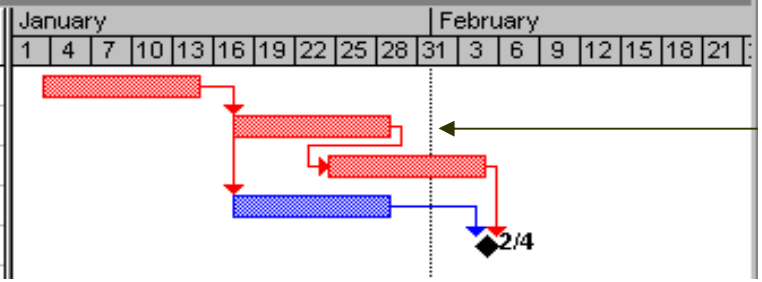
	Task Name	Duration	Start	Finish
1	A	10 days	Mon 1/3/00	Fri 1/14/00
2	B	10 days	Mon 1/17/00	Fri 1/28/00
3	C	5 days	Mon 1/31/00	Fri 2/4/00
4	D	10 days	Mon 1/17/00	Fri 1/28/00
5	E	0 days	Fri 2/4/00	Fri 2/4/00



Shortened duration

compress3.mpp

	Task Name	Duration	Start	Finish
1	A	10 days	Mon 1/3/00	Fri 1/14/00
2	B	10 days	Mon 1/17/00	Fri 1/28/00
3	C	10 days	Mon 1/24/00	Fri 2/4/00
4	D	10 days	Mon 1/17/00	Fri 1/28/00
5	E	0 days	Fri 2/4/00	Fri 2/4/00



Overlapped tasks

Importance of Updating Critical Path Data

- It is important to update project schedule information
- The critical path may change as you enter actual start and finish dates
- If you know the project completion date will slip, negotiate with the project sponsor

Program Evaluation and Review Technique (PERT)

- PERT is a network analysis technique used to estimate project duration when there is a high degree of uncertainty about the individual activity duration estimates
- PERT uses probabilistic time estimates based on using optimistic, most likely, and pessimistic estimates of activity durations

PERT Formula and Example

- **PERT weighted average formula:**

$$\frac{\textbf{optimistic time + 4X most likely time + pessimistic time}}{\textbf{6}}$$

- Example:

PERT weighted average =

$$\frac{\textbf{8}\text{ workdays} + 4 \text{ X } \textbf{10}\text{ workdays} + \textbf{24}\text{ workdays}}{6} = \textbf{12 days}$$

where 8 = optimistic time, 10 = most likely time, and 24 = pessimistic time